MILITARY SPECIFICATION

WELDING MACHINES, ARC, DC, TRANSFORMER-RECTIFIER, CONSTANT VOLTAGE, CONSTANT CURRENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers direct current, transformer-rectifier, welding machines supplying both the constant current and constant voltage arc characteristics.

1.2 Classification. The welding machines shall be of the following sizes as specified (see 6.2.1).

Size 300 - 300 Amperes
Size 600 - 600 Amperes

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletlonsl and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

NIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories , Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 Screw Thread Standards for Federal Services.

FED-STD-376 Preferred Metric Units for General Use by the Federal Government.

2,1.2 Other Goverment documents and publications. The followlng other Government documents and publications form a part of this specification to the extent spec~fled herein. Unless otherwise specified, the Issues shall be those In effect on the date of the solicitation.

U.S. DEPARTMENT OF LABCJR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001).

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity) .

2.2 Other publications. The followlng document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA EW 1 - Electrlc Arc-Weld~ng Apparatus.

(Application for copies should be addressed to the American National Standards Institute, ATTN : Sales Dept., 1430 Broadway, New York, NY 10018-3363).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187).

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available In or through llbrarles or other informational services).

'2.3 Order of precedence. In the event of a confllct between the text of this specification and the references cited herein (except) for associated detail specifications, specification sheets or MS standards) . the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding machines shall be new and one of the manufactures current models capable of operating as constant current or constant voltage machines. The machines shall include all components, parts and features necessary to meet the performance requirements specified herein.

3.2.1 Measurement system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (S1) shall be used in the design and construction of the welding machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification. all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to S1 units through the use of the conversion factors and methods speclfled In FED-STD-376.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials w1ll not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which w1ll restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and con-

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strutted for energy efficiency as governed by the developments available within the industry.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the weldlng machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection. maintenance , and repair of the parts. All machine parts, components . mechanisms , and assembles furnished on the machine, whether or not specifically required herein. shall comply with all requirements of OSHA 29 CFIt 1910 are applicable to the machine Itself.

3.2.5 Mercurv restriction. The welding machine shall not contain mercury or mercury compounds nor k< exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the weldlng machine.

3.2.7 Lubrlcatlon. Means shall be provided to ensure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.8 Interchangeability. To provide for replacement of worn parts , all parts shall be manufactured to definite dimensions and tolerances.

3.2.9 Open circuit voltage. The open circuit voltage shall be in accordance with NEMA EW-1. If open circuit voltage exceeds 80 volts, proof of less than 10 percent ripple voltage is required.

3,2.10 Temperature rise. The maximum allowable temperature rise of various Insulated parts of the machine for the given class of insulation system used shall be in accordance with NEMA EW-1.

3.2.11 Po~tabillty. The welding machines shall be portable, and shall be based mounted. Running gear. If required (see 6.2.1) , shall be as speclfled by the procuring activltv. The welding machine shall be provided with a llftlng device for handling by crane or hoist, and the device shall be capable of supporting 2-1/2 times the weight of the machine. The center of gravity of the entire unit shall be as low as is consistent with economical design. The machine shall also be designed to be llfted by forkllft truck.

3.3 Construction. The weldlng machine shall be constructed of parts which are new, without defects, and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to lts maximum rating and capacity without permanent distortion.

3.,3.1 Castings and forgings. A;l1 castings and forgings shall be free of defects, scale and mismatching. No processes such as welding, peen-

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ing, plugging, or fil}ing with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed In a manner to prevent change of tightness. Fastening devices subject to removal or ad~ustment shall not be swaged, peened , staked, or otherwise permanently Installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness 1s required to perform a necessary iunctlon. Except as otherwise specified herein, the condition and finish of all surfaces shall be In accordance with the manufacturer's commercial practice.

3.3.4 Welding brazing, or soldering. Welding, brazing or solderlng shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4 Components. The welding machine shall consist of, but not be limited to, the following components:

3.4.1 Enclosure. The welding machines shall be enclosed with a sheet metal enclosure in accordance with NEMA EW-1. The machines shall be mounted on a rigid base that will prevent distortion or misalignment of the machine or enciosure when placed on an uneven surface.

3.4.2 Transformer. The weldlng transformer shall be designed to draw a balanced three phase load from the input line. The primary and secondary coils shall be insulated from each other. Insulation of the transformer coils, shall be Class B, F, or H material in accordance with requirements of NEMA EW-1. The temperature rise of the wlndlngs shall not exceed the limits of the applicable class of insulation svstem used as specified in NEMA EW-1. Insulation distances and clearance shall be as liberal as 1s economically practicable in order to mlnlmlze the need for frequent cleaning. The transformer shall be rigidly mounted to the frame of the enclosure.

3.4.3 Rectifier. The rectifer shall be a three phase full wave type, Rectification shall be accomplished by silicon diodes of suitable size and design for the intended service. The rectifier shall be cooled by forced air. Temperature rise of the silicon diodes shall not exceed 90

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degrees C when operated at rated duty cycle.

3.4.4 Controls. The weldlng machines shail have all the necessary controls for both constant current and constant voltage operations. The controls and instruments shall consist baslcallv of a voltmeter and ammeter , on-off sw1tch , selector switch, slope and current control , voltage control, contactor control receptacle and receptacle for 115 volt AC, single phase power supply. All controls and meters shall be mounted on the front panel and so located for easy operation and observation.

3.4.4.1 Voltmeter and _ammeter. The welding machines shall have a voltmeter and an ammeter. The voltmeter shall indicate the weldlng machine's voltage at all times. and the ammeter shall Indicate the weld-lng current. 'rhe Instruments shall be flush mounted, panel type. and protected from dirt and moisture. The meters shall have an accuracy o!' ~ 2 percent of full scale reading.

3.4.4.'.2 On-off switch. The weldlng machines shall have a switch to turn the machines on or off to ellminate the need to disconnect the primary line. A control transformer, whose secondary voltage is used to maintain the control voltage on components mounted on the rear of the front panel at 115 volts or less, may remain energized, provided its secondary circuits are de-energized by the on-off switch.

3.4,4.3 Selector switch. The welding machine shall be provided with a selector switch to Indicate whether the machine is to be used as a constant current or constant voltage unit. When the switch is in the constant current position, the primary contactor will energize when power is applied and the on-off switch IS in the "on" position. When the selector switch IS in the constant voltage position and 115 volt AC is connected to the contactor receptacle. the contactor will be controlled by the switch on the gun, torch, or other remote control.

3.4.4.4 ~~e and current control. The weldlng machines shall have a slope and current control. When weldlng with constant current, the slope and current control shall vary the weldlng current output from minimum to maximum values of the machine. When welding with constant voltage, the slope and current control shall control the slope of the volt-ampere curve. The design of the machines shall be such that lt produces the proper transient characteristics for short circuiting globular, or spray transfer weldlng.

3.4.4.5 Volta~e control. The voltage for the constant mode shall be controlled by either a single continuous full range control or a voltage range switch. When the voltage range switch is provided, load adjust-ments within the ranges shall be made by a voltage or Inductance control.

3.4.4.6 Receptacles. The welding machine shall have a contactor receptacle and a 115 volt AC duplex receptacle.

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3.5 Size and characteristics. The welding machine shall have the minimum curredt range as shown in table I for the size machine.

3.6 Performance. When connected to a proper power source, the welding machines shall produce direct current. The welding machines shall be dual purpose units capable of provid~ng power for making' high qualltv welds on ferrous and non-ferrous metals using the manual stick electrode and automatic and semi-automatic processes. The arc characteristic . volt-ampere curves, shall be constant current and constant voltage types.

3.6.1 Constant current operation. The weldlng machines shall be capable of operating as constant current weldlng machines wherein manual stick electrode and other processes employing the drooping volt-ampere curve are employed.

3.6,2 Constant voltage operation. The welding machines shall be capable of operation as constant voltage welding machines f0s' use w~th automatic and semi-automatic welding sets. The load voltage shall be adjustable from a minimum range of 20 to 50 volts.

3,7 Marking on plates. All words on plates shall be In the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.8 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each welding machine and contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number of Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.9 Technical data. When technical data [operating manuals , maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) IS required, it shall be furnished in accordance with the requirements of DD Form 1423, all technical furnished shall be written in the Engllsh language.

3.10 Workmanship. Workmansh~p of the weldlng machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor 1s responsible for the performance of all inspection requirements specified herin. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facll~ties suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth In the speclflcatlon where such InspectIons are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Ftesponsiblllity for compliance. All i terns must meet all requirements of sections 3 and 5. The Inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows;

a. First article inspection (see 4.4).

b. Quallty conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), ail inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relatlve humidity, and the alr pressure existing inside the building at the time the lnspectlons, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection 1s required, lt shall be applled to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6,2.1), first article inspection shall consist of the examination in 4.6 and the tests in 4.7. Failure of the item to pass the first article examination or the tests shall be cause for re~ectlon.

4.5 Quallty conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quallty conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the Inspections shalI be cause for rejection.

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4.6 Examination. The welding machine shall be examined to determine compliance with all requirements of this specification includlng compliance with all referenced standards.

4.7 Tests.

4.7.1 High potential test. The welding machine shall be subjected to a high potential test as described In Testing Standard of NEMA-EW-1.

4.7.2 Rough handling test. When speclfled (see 6.2.1), the weldlng machine shall be subjected to the followlng rough handling test. A chain or cable shall be attached to the llftlng device of the weldlng machine. The welding machine shall be raised and then dropped to fall freely a distance between 6 inches minimum and 10 inches maximum upon a solld concrete slab. The welding machine shall withstand the shock without damage of any kind. This drop shall be repeated so that Impact is made on each of three lower edges of the machine. There shall be no damage to the welding machine as a result of this test, other than minor denting of the lower edges. This test shall be performed prior to all tests in 4.7.2 through 4.7.5.

4.7.3 Open circuit voltage test. The welding machine shall be operated on the specified primary voltage. The open circuit voltage range for constant current mode shall be in accordance with 3.6.2.

4.7.4 Performance test. The leads of the welding machine shall be connected to a resistance load bank that is rated in excess of the maximum potential of the welding machine. The welding machine shall be operated through successive periods of 10 minutes each in increments of 25, 50, 75 and 100 percent of rated output. The preceding test shall be applied to both modes of performance as specified In 3.6.1 and 3.6.2. The operating temperatures of the machine shall be recorded to determine compliance to 3.2.10. In addition to load tests, the following welding test shall be performed without any difficulty in establishing or maintaining the arc. To test the constant current performance of the machine, (six inch) weld beads shall be deposited using 1/8, 5/32 and 1/4 inch diameter stick electrodes with both straight and reverse polarlty. To test the constant voltage performance of the machine, six inch weld beads shall be deposited using the semi-automatic gas metal arc process with minumum and maximum current as shown In table I. All welding shall be done at minumum distance of 20 feet from the machine.

4.8 Inspection of packagin&. Packaging shall be inspected to determine conformance of the requirements of Section 5.

5. PACKAGING

5.1 Perservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

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6. NOTES

6.1 Intended use. The welding machines covered by this specification are Intended for any DC welding wherein the constant voltage and constant current volt-ampere characteristics are required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the followlng:

a. Title, number, and date of this specification.

b. S~ze required (see 1.2).

c. First article, lf required (see 3.1),

d. Power source, If difference (see 3.2).

e. If machine is required to be configured in a specific measurement system (US or S1) , state required system (see 3.2.1).

f. Portability, as specified (see 3.2.11).

13. Painting, if different (see 3.3.5).

h. Nameplate, if different (see 3.8).

1. Inspection conditions, If different (see 4.3).

j. First article inspection, if different (see 4.4).

k. Quallty conformance inspection, if different (see 4.5).

1. Rough handllng test, If requ~red (see 4.7.2).

m. If preservation, packing, and marking in accordance with MIL-W-45562 are required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manualS, parts lists. wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6,,3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under

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the contract., The pontractlng officer should Include speclflc instructions in acquisition documents regarding arran~ements for examination, test, and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used In this kevlsion to identify changes with respect to the prev~ous issue due to the extensiveness of the changes.

6.5 Subject term (keyword) listing.

Rectifier, three phase, full wave type
Slope and current control
Transformer, welding
Voltage control

TABLE I. Sizes and Characteristics.

*current range (amperes)		
Size	Constant voltage	Constant current
300	90-300	40-300
600	90-600	90-500

*constant voltage rated at 100X duty cycle.
Constant current rated at 60% duty cycle.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - SH
Air Force - 84

User activities:
Navy - MC, CG, AS, SH

Preparing activltv:
DLA - 1P

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